Physics 438A – Chapter #9
The Hydrogen Atom

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1 The Electrostatic Model

In this chapter, we use the *electrostatic model* for the hydrogen atom, in which the electron and proton interact by an electrostatic force determined by the instantaneous positions of the particles. The electrostatic model is an approximation to electrodynamics in which one neglects the delay in the communication between the particles, magnetic effects, and the effects of radiation. As a part of the electrostatic model we shall also assume the electron motion is non-relativistic, something that is justified in the case of hydrogen.

Under these assumptions, the potential energy of interaction between the electron and the proton is

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r} = -\frac{k_e e^2}{r}$$

and the corresponding radial equation is

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2} + \left[\frac{\hbar^2}{2\mu}\frac{\ell(\ell+1)}{r^2} - \frac{k_e e^2}{r}\right]u = Eu$$

where μ is the reduced mass of the electron-proton system

$$\mu = \frac{m_e m_p}{m_e + m_p}$$

To a good approximation, we could take $\mu = m_e$ since the proton is much heavier than the electron. In fact, the ratio of the proton mass to the electron mass is roughly 1836, which is suspiciously close to the pure number $6\pi^5$. The reduced mass differs from the mass of the electron by about 5 parts in 10^4, which is within the limits of experimental precision.

The Coulomb potential admits continuum states (in this case with $E > 0$) and discrete states (with $E < 0$). The Hamiltonian therefore has a mixed spectrum, which implies the bound states are not complete on their own. For now, we are only interested in the bound states since they correspond to a stable hydrogen atom. We'll begin by solving the radial equation, and along the way, we'll discover the bound state energies match the semi-classical Bohr model you may have learned about in a previous class. This chapter will conclude with a discussion of the hydrogen wave functions.

2 The Radial Equation

It is convenient to work with natural units after we identify the relevant physical scales of the hydrogen atom. This may be accomplished either by dimensional analysis or by carefully inspecting the radial equation. Natural units will also make it easier to simulate the hydrogen atom with a computer. The radial equation may be written as

$$\frac{d^2u}{dr^2} + \left[\frac{2\mu E}{\hbar^2} - \frac{\ell(\ell+1)}{r^2} + \frac{2\mu k_e e^2}{\hbar^2}\frac{1}{r}\right] u = 0$$

Each term in the brackets above has units of 1/length^2, so let's define a characteristic distance based on the physical quantities that appear in the third term specifically:

$$a \equiv \frac{\hbar^2}{2\mu k_e e^2}$$

The distance a is very close to the Bohr radius which is 0.529 Å. Let's use natural units where $\hbar = a = 2\mu = 1$. The energy and momentum scales are then given by

$$[\text{energy}] = \frac{\hbar^2}{2\mu a^2}, \qquad [\text{momentum}] = \frac{\hbar}{a}$$

Lastly, since $E < 0$, we can define a positive constant γ such that

$$\gamma \equiv \sqrt{-E} \quad \Rightarrow \quad E = -\gamma^2$$

In terms of dimensionless parameters, the radial equation becomes

$$\frac{d^2u}{dr^2} + \left[\frac{2}{r} - \frac{\ell(\ell+1)}{r^2} + \gamma^2\right] u = 0$$

Asymptotic Solutions

Figuring out solutions for large and small r is interesting in its own right, and our knowledge of the asymptotic solutions may allow us simplify the differential equation. We also need to ensure that our solutions are normalizable; an asymptotic solution is the first place to look for any pathological behavior. For large r, the radial equation is approximately

$$\frac{d^2u}{dr^2} = \gamma^2 u$$

The general solution for large r is $u = Ae^{-\gamma r} + Be^{\gamma r}$, but the wave function must vanish at $r \to \infty$, so we must choose $B = 0$. For large r, we find $u = Ae^{-\gamma r}$. Alternatively, when r is small, the centrifugal term dominates, and the radial equation in this limit is

$$\frac{d^2u}{dr^2} = \frac{\ell(\ell+1)}{r^2}u$$

Monomial functions have the property that $(r^s)' = sr^{s-1}$, so with $u = r^s$,

$$s(s-1)u^{s-2} = \ell(\ell+1)u^{s-2} \to s(s-2) = \ell(\ell+1)$$

The general solution for small r is $u = Cr^{\ell+1} + Dr^{-\ell}$. The second term grows without bound for small r (assuming $\ell > 0$), so we must choose $D = 0$. The solution for small r is then $u = Cr^{\ell+1}$. Now that we know the asymptotic solutions, we can find the solution for intermediate r by solving for a function $v(r)$ such that

$$u(r) = r^{\ell+1}e^{-\gamma r}v(r) \quad \text{or} \quad \mathcal{R}(r) = r^{\ell}e^{-\gamma r}v(r)$$

After substituting $u(r)$ as written above into the differential equation (this was quite tedious; you should take moment to make sure it's correct),

$$r\frac{d^2v}{dr^2} + 2(1+\ell-\gamma r)\frac{dv}{dr} + 2(1-\gamma-\gamma\ell)v = 0$$

Series Solution Method

We will use a power series expansion to solve the radial equation for $v(r)$. Assume

$$v(r) = \sum_{k=0}^{\infty} c_k r^k$$

where c_k are coefficients to be determined. While this approach is powerful, it demands caution. One must take care when using the series solution method. The differential equation we aim to solve is *stubborn*. For large r,

$$\frac{d^2v}{dr^2} - 2\gamma\frac{dv}{dr} = 0$$

The non-trivial solution to this asymptotic equation is $e^{2\gamma r}$, and furthermore,

$$u(r) = r^{\ell+1}e^{-\gamma r}e^{2\gamma r} \propto e^{\gamma r}$$

It seems like the solution really wants to grow without bound for large r, and we must eliminate this possibility on physical grounds. If we require the series to terminate for a finite value of the index, we can remove the exponential solution.

After substituting the series solution,

$$\sum_k k(k+1)c_{k+1}r^k + 2(\ell+1)\sum_k (k+1)c_{k+1}r^k \\ -2\gamma\sum_k kc_kr^k + 2(1-\gamma-\gamma\ell)\sum_k c_kr^k = 0$$

In order for all terms of the series to sum to zero for any and all values of r, the coefficient of each power of r must vanish. Hence,

$$k(k+1)c_{k+1} + 2(\ell+1)(k+1)c_{k+1} - 2\gamma kc_k + 2(1-\gamma-\gamma\ell)c_k = 0$$

$$c_{k+1} = \frac{2\gamma(k+\ell+1)-2}{(k+1)(k+2\ell+2)}c_k$$

This is a recurrence relation for the coefficients; the starting coefficient c_0 determines all of the remaining coefficients in the function $v(r)$. Ultimately, c_0 is determined by normalization.

Termination Condition

Notice that for large values of k,

$$c_{k+1} \approx \frac{2\gamma}{k+1}c_k \Rightarrow c_k = \frac{(2\gamma)^k}{k!}$$

which is the same recurrence relation for $e^{2\gamma r}$. There must exist $k_{\max}$ such that $c_{k_{\max}+1} = 0$. From the recurrence relation, we can see that the coefficients will vanish if

$$\gamma(k_{\max}+\ell+1) - 1 = 0$$

Since $k_{\max}$ and ℓ are integers, it follows that

$$n = k_{\max}+\ell+1 = 1, 2, 3, \ldots$$

is also an integer, and by the termination condition,

$$\gamma = \frac{1}{n}$$

Enforcing the boundary conditions on the radial wave function has lead to quantization of energy! More precisely, we find that

$$E = -\gamma^2 = -\frac{1}{n^2} \quad \text{(natural units)}$$

$$= -\frac{\hbar^2}{2\mu a^2}\frac{1}{n^2} \qquad \text{(ordinary units)}$$

The boundary conditions also constrain the values of ℓ. For a given n, $\ell = n - k_{\text{max}} - 1$ has a maximum value when $k_{\text{max}} = 0$, so $\ell_{\text{max}} = n - 1$. We now have all the "quantum numbers" of the hydrogen atom! Let's summarize what we know so far:

1. **Principal quantum number:** $n = 1, 2, 3, \ldots, \infty$ (also called the shell number)
2. **Angular momentum quantum number:** $\ell = 0, 1, 2, \ldots n - 1$
3. **Magnetic quantum number:** $m = -\ell, -\ell + 1, \ldots, 0, \ldots, \ell - 1, \ell$

The magnetic quantum number is named after its relevance to the splitting of spectral lines when the atom interacts with a magnetic field.

3 Hydrogen Energies and Spectra

Approximating the reduced mass as the mass of the electron alone, we arrive at the following set of discrete energies labeled by the principle quantum number:

$$E_n = -\frac{\hbar^2}{2\mu a^2}\frac{1}{n^2} = -\frac{m_e c^2}{2n^2}\left(\frac{k_e e^2}{\hbar c}\right)^2 = -\frac{\alpha^2}{2n^2} m_e c^2$$

where $\alpha = 1/137$ is the **fine structure constant**. In this form, it is easy to see that the result has units of energy, roughly on the order of 10 eV. Somewhat more precisely,

$$\boxed{E_n = -\frac{13.6 \text{ eV}}{n^2} = -\frac{1}{n^2}\text{Ryd}}$$

where Ryd stands for "Rydberg." Here are some important features of the hydrogen energies:

1. There are an infinite number of bound states.
2. The energy levels are degenerate. For each energy E_n, there are n possible ℓ states and for each ℓ, there are $2\ell + 1$ possible m states. The total number of states at each

energy level E_n is the sum of these possibilities:

$$\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$$

If we include the spin of the electron, the total number is $2n^2$. The m-degeneracy is related to spherical symmetry while ℓ-degeneracy is related to a special dynamical symmetry of the $1/r$ potential (Runge-Lenz symmetry). Roughly speaking, m-degeneracy reflects no preferred direction in space, while the ℓ-degeneracy reflects the fact that in a $1/r$ potential, different orbital shapes can have the same energy.

Hydrogen atoms absorb or emit light when electrons transition between energy levels. The photon energy always matches the energy difference:

$$E_{\text{photon}} = \Delta E = |E_f - E_i|$$

The energy of a photon is related to its wavelength:

$$E_{\text{photon}} = \hbar\omega = hf = \frac{hc}{\lambda}$$

Hence, the wavelengths of light that can be emitted or absorbed by hydrogen are given by

$$\boxed{\frac{1}{\lambda} = R_\infty \left| \frac{1}{n_i^2} - \frac{1}{n_f^2} \right|}$$

where

$$R_\infty = \frac{\alpha^2 m_e c^2}{4\pi\hbar c} = \frac{m_e e^4}{8\varepsilon_0 h^2 c} \approx 10.97 \times 10^6 \text{ m}^{-1}$$

Not all transitions are allowed. Conservation of angular momentum leads to the following **selection rules:**

$$\Delta\ell = \ell_f - \ell_i = \pm 1$$

$$\Delta m = m_f - m_i = 0, \pm 1$$

The photon itself has spin angular momentum (helicity) of 1, so the atom's angular momentum must change as a result of absorption or emission of a single photon.

4 Radial Wave Function

The radial wave function with dimensions back in place is

$$\mathcal{R}_{n\ell}(r) = \left(\frac{r}{a}\right)^{\ell} e^{-r/na} v(r/a)$$

where

$$v(r/a) = \sum_{k=0}^{n-\ell-1} c_k (r/a)^k$$

and

$$c_{k+1} = \frac{2\gamma(k+\ell+1)-2}{(k+1)(k+2\ell+1)} c_k$$

Example: Ground State

The state with lowest energy, the ground state of the hydrogen atom, is labeled by $n = 1$ and $\ell = 0$. The ground state radial wave function is

$$\mathcal{R}_{10}(r) = c_0 e^{r/a}$$

where c_0 is determined by normalization of the radial wave function:

$$1 = \int_0^{\infty} dr\, r^2 |\mathcal{R}_{10}(r)|^2 = \int_0^{\infty} dr\, r^2 |c_0|^2 e^{-2r/a} = |c_0|^2 \cdot \frac{a^3}{4}$$

The full ground state radial wave function can be written as

$$\mathcal{R}_{10}(r) = \frac{2}{a^{3/2}} e^{-r/a}$$

Example: Excited States

The following states all have the same energy:

$$|n, \ell, m\rangle = \left\{ |2,0,0\rangle, |2,1,-1\rangle, |2,1,0\rangle |2,1,1\rangle \right\}$$

For the state $|2,0,0\rangle$, $k_{\max} = 1$, and $v(r/a) = c_0 + c_1(r/a)$. By the recursion relation,

$$c_1 = -\frac{1}{2} c_0$$

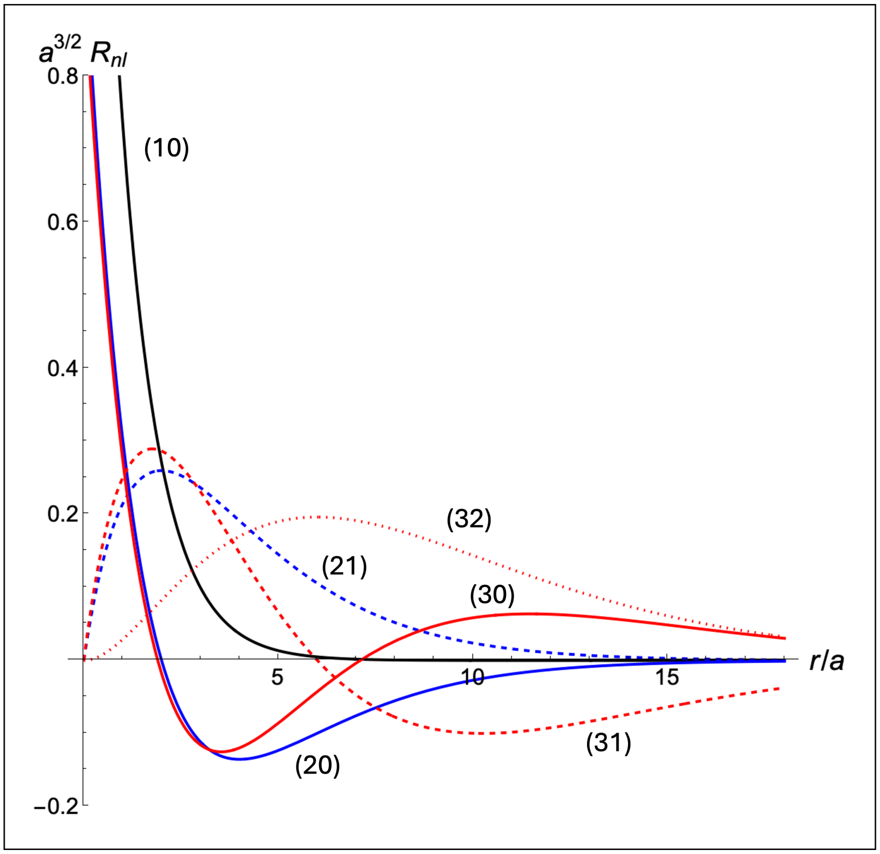

Figure 4.1: Radial wave functions for the ground state (10) and the first few excited states.

so the radial wave function is

$$\mathcal{R}_{20}(r) = c_0 e^{-r/2a}(1 - r/2a)$$

For states with $n = 2$ and $\ell = 1$, $k_{\text{max}} = 0$, so

$$\mathcal{R}_{21}(r) = c_0 r e^{-r/2a}$$

Note that in each of the expressions above, the term c_0 is particular to that solution–they are not the same, but they are all determined by normalization.

The radial wave functions can be written in terms of special functions called associated Laguerre polynomials $L_q^p(x)$, where q indicates the degree of the polynomial. In terms of associated Laguerre polynomials, the radial wave functions are

$$\boxed{\mathcal{R}_{n\ell}(r) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-\ell-1)!}{2n(n+\ell)!}} \left(\frac{2r}{na}\right)^\ell \left[L_{n-\ell-1}^{2\ell+1}(2r/na)\right] e^{-r/na}}$$

where

$$L_q^p(x) = \frac{x^{-p}e^x}{q!}\frac{d^q}{dx^q}\left(e^{-x}x^{q+p}\right) = (-1)^p\frac{d^p}{dx^p}L_{p+q}(x)$$

and $L_q(x)$ are the ordinary Laguerre polynomials defined as

$$L_q(x) = \frac{e^x}{q!}\frac{d^q}{dx^q}\left(e^{-x}x^q\right)$$

The radial wave function $R_{n\ell}$ has $n-\ell-1$ nodes and $n-\ell$ antinodes. For a given n, $R_{n\ell}$ has fewer nodes for higher ℓ states, and the angular wave functions "compensate" by having more nodes. See Figure 4.1 for a visualization of the radial wave functions.

5 Full Hydrogen Wave Functions

We can recombine the three separated parts of the wave function to form the full three-dimensional energy eigenstate wave functions of the hydrogen atom:

$$\boxed{\psi_{n\ell m}(r,\theta,\phi) = \mathcal{R}_{n\ell}(r)Y_\ell^m(\theta,\phi)}$$

These states are also eigenstates of the angular momentum operators L^2 and L_z; they are simultaneous eigenstates of H, L^2, and L_z altogether. The three eigenvalue equations are

$$H\psi_{n\ell m}(r,\theta,\phi) = -\frac{13.6\text{ eV}}{n^2}\,\psi_{n\ell m}(r,\theta,\phi)$$

$$L^2\psi_{n\ell m}(r,\theta,\phi) = \ell(\ell+1)\hbar^2\,\psi_{n\ell m}(r,\theta,\phi)$$

$$L_z\psi_{n\ell m}(r,\theta,\phi) = m\hbar\,\psi_{n\ell m}(r,\theta,\phi)$$

The normalization condition for the full wave function is the three-dimensional integral

$$\begin{aligned} 1 &= \int d^3r\,|\psi_{n\ell m}(r,\theta,\phi|^2 \\ &= \left\{\int_0^\infty dr\,r^2|\mathcal{R}_{n\ell}(r)|^2\right\}\left\{\int_0^\pi\int_0^{\pi/2} d\theta d\phi\,\sin\theta|Y_\ell^m(\theta,\phi)|^2\right\} \end{aligned}$$

See Figure 5.1 for a static visualization of the hydrogen wave functions.

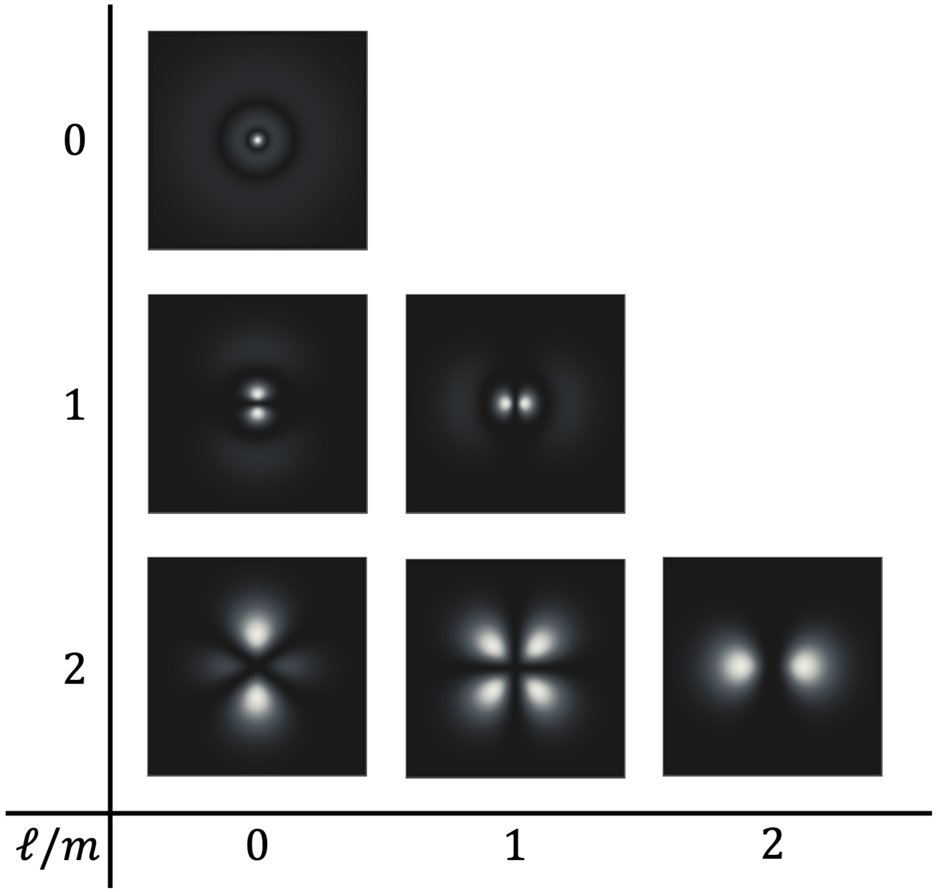

Figure 5.1: Hydrogen atom probability densities $|\psi_{n\ell m}|^2$ for $n = 3$, shown in the xz-plane ($y = 0$) passing through the nucleus at the origin. Rows correspond to $\ell = 0, 1, 2$ (top to bottom) and rows to $m = 0, 1, 2$ (left to right where allowed). Bright regions indicate high probability density. Radial nodes (spherical shells) are given by $n-\ell-1$: the $3s$ state ($\ell = 0$) has 2 radial nodes, $3p$ ($\ell = 1$) has 1 radial node, and $3d$ ($\ell = 2$) has no radial nodes. Angular nodes are determined by ℓ: 0 for s, 1 for p (a nodal plane), and 2 for d states (combinations of nodal planes/cones), visible as directions where the density vanishes.